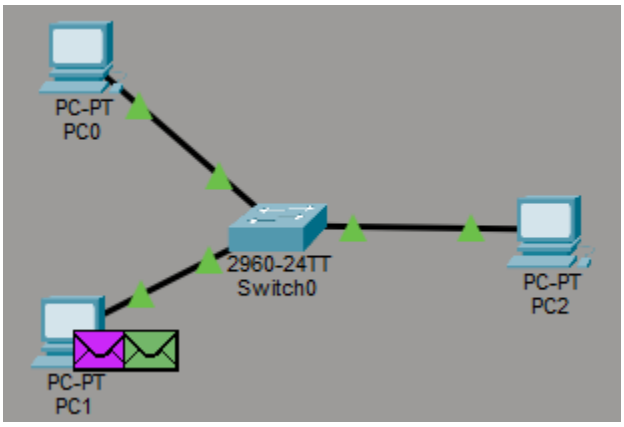
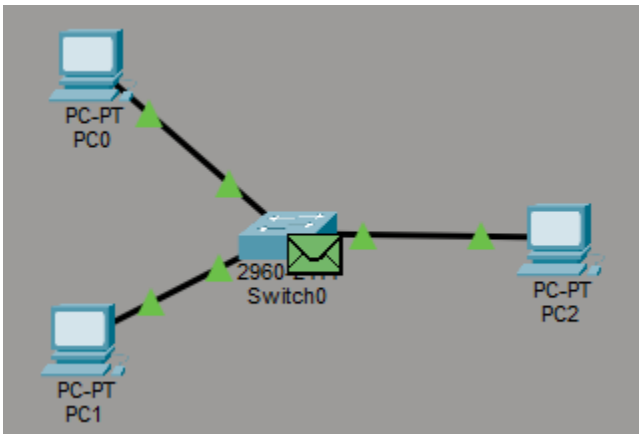
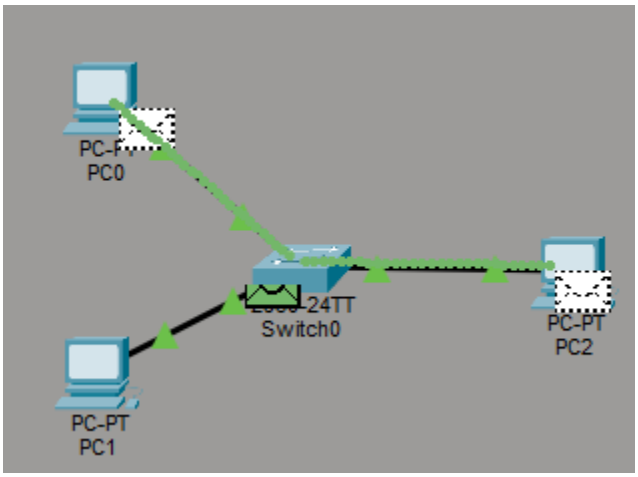
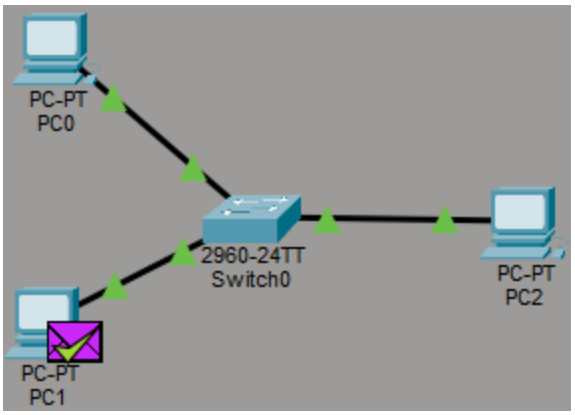
1)


PC-PT
PC0
2960-24TT
Switch0
PC-PT
PC2
PC-PT
PC1


PC-PT
PC0
Switch0
PC-PT
PC2
PC-PT
PC1


PC0
Switch0
PC-PT
PC2
PC-PT
PC1

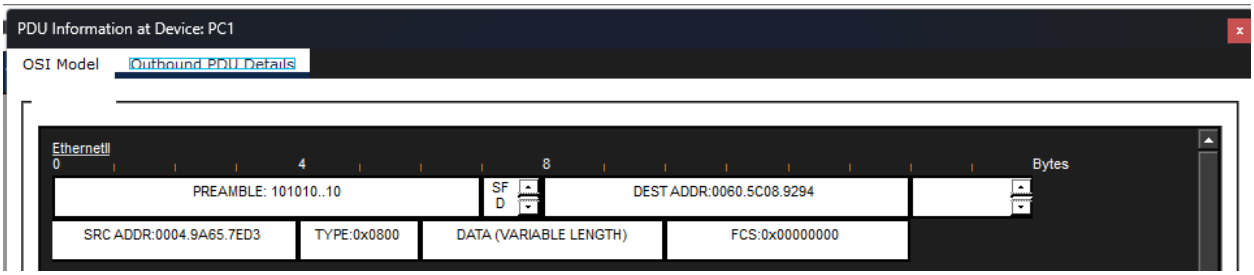



```
Switch#enable
Switch#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0002.1718.e4d3    DYNAMIC     Fa0/1
   1    0004.9a65.7ed3    DYNAMIC     Fa0/2
   1    0060.5c08.9294    DYNAMIC     Fa0/3
Switch#
```

2)

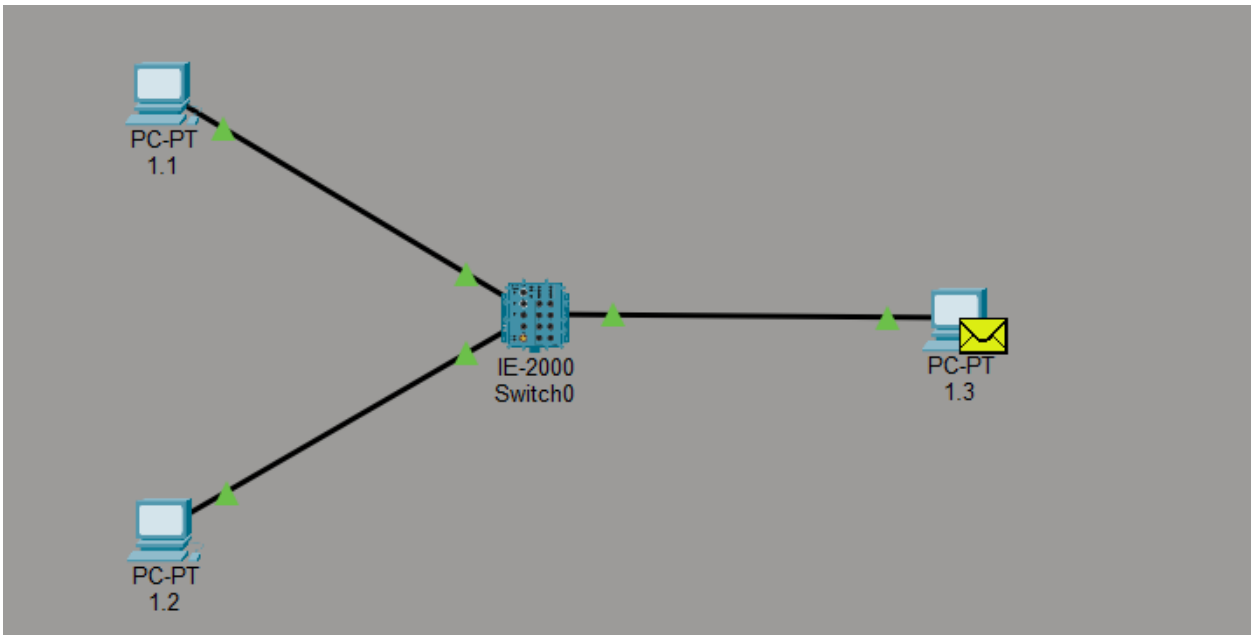


3)PING

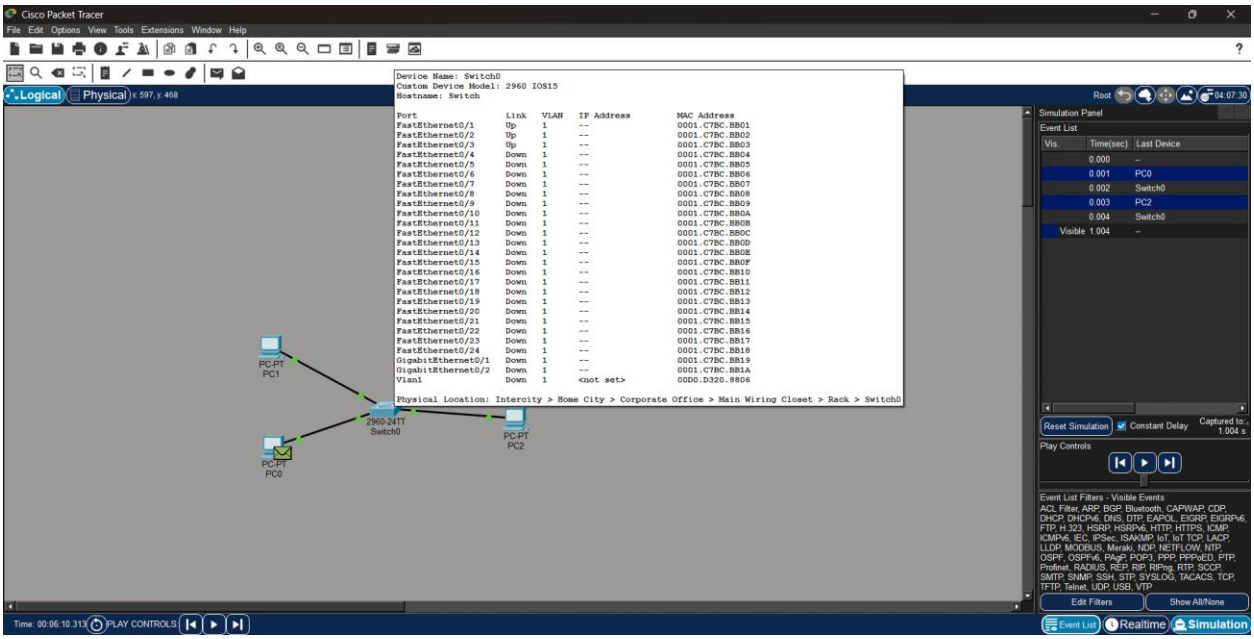
```
Switch#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0009.7c39.80c2    DYNAMIC     Fa1/3
   1    00d0.bc42.e736    DYNAMIC     Fa1/1
Switch#
```

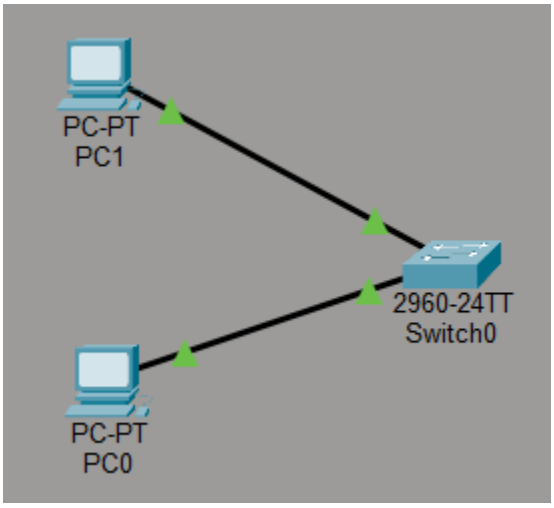


AFTER CHANGING MAC ADDR



4)

LAN SETUP



PING

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=8ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 4ms, Maximum = 8ms, Average = 5ms

LINUX KERNAL HANDLING OF ETHERNET DEVICES

The Linux kernel handles Ethernet communication through a structured networking stack.

Key Components:

- **NIC Driver**: Controls hardware (e.g., Ethernet card)
- **net_device structure**: Abstract representation of network interface

- **Socket Buffer (sk_buff)**: Data structure for packet handling

Packet Flow:

1. Packet received by NIC
2. Interrupt triggered → handled by driver
3. Packet stored in **sk_buff**
4. Passed through layers:
    a. Data Link → Network → Transport
5. Delivered to application

Mac Address Table of a Switch

```
Switch>show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.64be.d316    DYNAMIC     Fa0/1
   1    00e0.f98c.1b12    DYNAMIC     Fa0/2
Switch>
```

Conclusion:

Ethernet communication in a LAN is achieved through ARP-based address resolution and MAC-based frame forwarding, while the Linux kernel efficiently manages packet processing through its networking stack.P